order30 RESULTS
PREREG sha256 f599d527d3942fbf75466c87f83e6719334ac2d0721cf493ff57cb62dc79b570 (hashed before numerics).
Run note: the first run was interrupted after depth 1 (partial log logs/o30_run1_interrupted.log, kept). The unchanged script was rerun in full (logs/o30_run2.log); its depth-1 numbers are identical.
Controls (all PASS)
C1 corner(K) = CO: 0.0. C2 ‖R(CO)‖ = 54.406: rel err 0.0. C3 subdivision sum: ≤ 2.0e−13 (n ≤ 4). C4 Σ™ = L: ≤ 1.3e−14. arccosh(√(7/2)) = ½log(6+√35): 0.0.
RG23 (primary): arc-length weights W_n = Σ (™_i/L) Cor_{C_i}: FAIL
n	cones	d (non-vac)	‖R(W_n)‖	e_n = ‖R(CO)+R(W_n)‖/‖R(CO)‖	|cos|	best-scale s_n (λ)	near-law of W_n (κ free)
1	2	0.188	37.2	1.499	0.571	0.821 (−0.84)	0.0577, κ = 0.998
2	4	0.275	46.7	1.496	0.293	0.956 (−0.34)	0.0783, κ = 0.998
3	8	0.326	50.0	1.505	0.229	0.973 (−0.25)	0.0842, κ = 0.995
4	16	0.305	45.9	1.422	0.185	0.983 (−0.22)	0.0777, κ = 0.999

e_4 = 1.42 ≥ 0.1 and c_4 = 0.18 ≤ 0.99, so FAIL. The divergence flag is not raised: ‖R(W_n)‖ stays bounded at about 46–50, unlike order29's Alt_n. The arc-length rule converges in size but does not cancel R(CO); R(W_n) is nearly orthogonal to R(CO) and increasingly so. Prediction FAIL correct (blind).
RG24 (secondary): uniform average U_n: FAIL by rule, non-discriminating
t_n = 0.490, 0.735, 0.857, 0.918 (n = 1..4); d(CO − U_n) = 0.0069 (> 1e−3, passes the written threshold).
Disclosure (post-hoc): by the subdivision identity, U_n = 2^{−n}(CO + ¼(2ⁿ−1)vtx) exactly. So CO − U_n = (1−2^{−n})(CO − ¼vtx) lies in span{CO, vtx}, and t_n → ‖R(CO − ¼vtx)‖/‖R(CO)‖. RG24 carries no information beyond CO and the constant vertex vector. The PREREG non-vacuity span should have included vtx (lesson recorded).
Near-law residual of U_n: 0.062, 0.072, 0.105, 0.182 (κ 0.998 → 0.975).
Descriptive finding (Category B)
Every corner object tested (CO, W_1..W_4) shows the same near-law, with κ = 0.995–0.999 and residual 0.058–0.084, including objects far from span{CO, Γ_γCO} (d up to 0.33). The near-law therefore looks generic to unimodular cone corners and not specific to K. Not preregistered; to be tested as a hashed map.
Answers
No combination cancels R(CO) under the arc-length rule. Category A: none new. v11: no. Ξ: not identified. E1–E3 OPEN.

order30 NOTES
- Arc-length weights give bounded residuals with fading anti-alignment (cos 0.57 → 0.18). Cancellation of R(CO) by subdivision-corner combinations with geometric weights is not supported (orders 28–30).
- Lesson: the non-vacuity span must be span{CO, Γ_γCO, vtx}; uniform subdivision averages are affine in CO.
- Open next step (order31, not started): preregister RG25. Hypothesis: the near-law (κ ≈ 1, r ≈ 0.06–0.08) holds for each single unimodular-cone corner Cor_C, with residual controlled by the arc length ™(C). Test the single cones from the depth-4 Stern–Brocot set (already in data/o30_cones.npz) with fixed thresholds, and look for an exact geometric law for the per-cone defect R(Cor_C) − R_ref.
- Naming: per Arvin's rule, labels L1–L7 are reserved (Einstein–Hampton 11D theorems). Order30 uses none. The existing order27 PREREG/RESULTS (already written) use "L1–L3" as local claim tags, but the paper v10 uses Lemma "parity" (no L-number). They were not edited (new files only); future orders use RG*/Lemma V-style names.
- Geometric only; MAPS_CANDIDATES (BATCH8, DISCOVER, MARKOV, L3) untouched; no Resonant objects; no bridge claim; CORE_FREEZE. nice, OMP_NUM_THREADS=1, 3 GB; no push or sends; new files only.